

Multiple Choice (10 marks)

1. The polynomial which results from the expansion of $(x^2+5x+6)^2+(px+q)(x^3+7x^2+3x)$ has degree 2. Find $p+q$.
 - (a) -2
 - (b) -4
 - (c) -1
 - (d) 1
2. How many numbers are in the list $25, 26, 27, \dots, 99, 100$?
 - (a) 75
 - (b) 76
 - (c) 22
 - (d) 23
3. The length of a rectangle is twice its width. Given the length of the diagonal is $5\sqrt{5}$, find the area of the rectangle.
 - (a) 2500
 - (b) 2
 - (c) 50
 - (d) 25
4. Find the remainder when 25^{1059} is divided by 23.
 - (a) 11
 - (b) 13
 - (c) 8
 - (d) 5
5. The sum $1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = n(n+1)(2n+1) \div 6$. What is the value of $21^2 + 22^2 + \dots + 40^2$?
 - (a) 41
 - (b) 22140
 - (c) 19270
 - (d) 2870

True / False (10 marks)

Answer True or False

6. True or False: The total number of liters of water in all 8 fish tanks is 280 liters.
7. True or False: The solution to the equation $-2k = -34.8$ is $k = 17.4$.
8. True or False: The area of a square with side length equal to the radius of a circle with a circumference of 12 is 12 square units.
9. True or False: 49 students on the dance team take tap lessons, which is more than half of the total 72 students.
10. True or False: Sapphire can make 25 bouquets by using 7 balloons in each bouquet from her total of 179 balloons.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Marshall has 550 words left to type for his English paper, and he has 25 minutes until class starts. Calculate the minimum typing speed required and discuss how you arrived at your answer.
12. Explain the relationship between the measurement of angles and the division of a circle, specifically discussing how many degrees are in an angle that represents $\frac{1}{360}$ of a full circle.

End of Paper